LIN CHENGDE

AI-Native Platform Engineer

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EXPERIENCE

Senior Software Engineer

ASUS Intelligent Cloud Service (AICS)

Nov 2023 – Ongoing | Singapore

Led the AI-native transformation of the xHIS clinical platform — architecting the micro-frontend infrastructure, agentic development tooling, and knowledge systems that power it.

- Spearheaded an **AI-native development ecosystem**: authored **Agent Skills** and engineered **20+ MCP tools** spanning package intelligence (component discovery, interface constraints), workflow integration (ticket ingestion, spec diffing, template retrieval), and design-system-aware context injection — enabling an agentic loop where **non-technical designers prompt-generate deployable widgets**, reducing designer UI iteration effort by ~**75%** and eliminating handoff overhead.
- **Co-invented two Taiwan IPO patents**: **xHIS Data Store (xDS)**, a shell-layer clinically-structured state store with multi-version schema tolerance enforcing real-time patient identity consistency across micro-frontend widgets — **eliminating patient data contamination risk**; and **xEmulator**, a cloud-native simulation environment for isolated cross-team widget development and integration testing.
- Engineered a **Schema-Driven Development** pipeline auto-generating DB schemas, TypeScript interfaces, and a **centralized frontend data access layer** from a single source of truth — compressing a 2–3 day multi-engineer cross-language sync process to **under 10 minutes**, with end-to-end type safety across the full stack.
- Architected a **micro-frontend platform** leveraging **Vite** and **Native ESM** with `vite-plugin-federation`, decoupling the monolithic xHIS application into autonomous, independently deployable widgets — boosting feature delivery speed by **46%** and reducing defect age by **58%**.
- Built a **multi-source RAG support agent** ingesting conversational logs, support tickets, PRDs, and codebase — achieving **91% recall** on benchmarked xHIS operational queries.

Software Engineer

Asus Intelligent Cloud Service (AICS)

Sep 2021 – Oct 2023 | Taiwan

- Designed integration abstractions to unify heterogeneous hospital legacy APIs, enabling scalable multi-institution onboarding; built E2E and stress testing frameworks achieving **100%** critical defect coverage pre-release.
- Built an ETL pipeline for medical time-series data into Elasticsearch, reducing query latency by **90%**; devised a partial document regeneration strategy cutting update time by **80%+**.

SKILLS

Languages TypeScript | Python | HTML/CSS | C/C++

Frameworks Vue3 | Vite | LangGraph | Express | Electron

Databases MongoDB | PostgreSQL | Redis | Elasticsearch

DevOps Docker | K8s | Azure | AWS | Grafana

AI EXPERTISE

Agentic Systems

Multi-Agent / A2A | LangGraph | Agent Skills | Claude Code

Protocol & Retrieval

MCP | Multi-source RAG | Agentic Retrieval

Reliability

AI Observability | Context Engineering | Harness Engineering

ACHIEVEMENTS

Dept. Code Champion Award
Awarded for consistently delivering high-impact, high-quality code, recognized by senior leadership.

Dept. Peer Feedback Award
Voted by peers for outstanding collaboration and cross-team impact.

Host of GenAI Application Seminars
Hosted internal seminars on generative AI in product innovation and workflow optimization.

Outstanding Performance
Fast-tracked to Senior Software Engineer in 2 years (vs. 4–5 avg) for outstanding performance.

EDUCATION

M.Sc. in Computer Science & Information Engineering (AI Program)

National Taiwan University

Sep 2019 – July 2021

GPA: 4.24 / 4.30 Rank: **1st in Dept.**

B.Sc. in Physics, Minor in Computer Science

National Taiwan Normal University

Sep 2015 – June 2019

Valedictorian, College of Science